National University of Singapore

School of Computing
Quiz 2 for
Semester 2 AY25/26

CS4221/CS5421 Database Application Design and Tuning

March 2026 Time Allowed 60 minutes

INSTRUCTIONS TO CANDIDATES

1. Please read all the instructions carefully. You are **NOT** allowed to start writing until you are told to do so.
2. This assessment paper contains **15 questions** and comprises **10 printed pages**, including this page. The last **2 page(s) are left empty** for your working.
3. Write all your answers in the provided space in provided the answer sheet. Answers not in the provided space will not be graded.
4. Erase any unwanted answers completely. Anything not erased completely will be treated as part of the answer. If a dashed line is provided, each dashed line can only be used for one line.
5. Write legibly. Marks may be deducted for illegible writing.
6. The total marks for this assessment is **50 points**. The mark for each question or sub-question is given in the question.
7. This is a **CLOSED-BOOK** assessment. You are only allowed to refer to **one double-sided A4 cheatsheet**. No electronic devices are allowed including but not limited to calculator, smart-watches, etc.
8. All SQL queries are to be executed on PostgreSQL 16.

This paper is a **practice paper**.
While it is intended to be similar to the actual assessments, no guarantee can be made to the degree of similarity.
Other potential topic includes planning/tuning on single table, transforming OLTP to OLAP, etc.

GOOD LUCK!

Your company, **Apasaja Private Limited**, is commissioned by an online shipping company *KirimBarang* to design a relational schema to manage their users, items, and orders. They currently use the following table. We will be using the letter alias for our analysis.

Email	**Name**	**Address**	**Country**	**ItemID**	**Price**	**OrderID**
A	B	C	D	E	F	G

There are the following constraints identified by *KirimBarang*.

1. A user is identified by email address. A user information also contains the name and address.
2. As they are shipping to a house, the address also uniquely identify a user.
3. Additionally, the address unambiguously identifies the country in which it is located.
4. An item is identified by item id. An item information also contains the price of the item.
5. Every order has an order id. If a user buys an item, given the combination of email and item id, we can know the order id.

Let $R = A, B, C, D, E, F, G$ and let Σ be the set of functional dependencies obtained by translating the 5 constraints above into a functional dependency constraints.

Multiple Response Questions (10 points)

For each question below, **shade your answer in the corresponding bubble on the answer sheet.** If multiple answers are equally appropriate, pick one and shade the chosen answer on the answer sheet. Do **NOT** shade more than one answer. If none of the answers is appropriate, shade X in the answer sheet.

1. (2 points) Which of the following functional dependencies are logically entailed by Σ. Select all options that apply.

   A. $\{A, B\} \rightarrow \{C\}$

   B. $\{E, F\} \rightarrow \{G\}$

   C. $\{B, D\} \rightarrow \{A\}$

   D. $\{C\} \rightarrow \{A, B, D\}$

   Shade X in the answer sheet if none of the options are true.

2. (2 points) Which of the following set of attributes is a candidate key of R with Σ? Select all options that apply.

   A. $\{A, E\}$

   B. $\{B, E\}$

   C. $\{C, E\}$

   D. $\{D, E\}$

   Shade X in the answer sheet if none of the options are true.

3. (2 points) Which of the following functional dependency of the form $X \rightarrow \{A\}$ violates the BCNF property of R with Σ? Select all options that apply.

   A. $\{A\} \rightarrow \{B\}$ B. $\{C\} \rightarrow \{A\}$ C. $\{E\} \rightarrow \{F\}$ D. $\{A, E\} \rightarrow \{G\}$

   Shade X in the answer sheet if none of the options are true.

4. (2 points) Which of the following functional dependency of the form $X \rightarrow \{A\}$ violates the 3NF property of R with Σ? Select all options that apply.

   A. $\{A\} \rightarrow \{B\}$ B. $\{C\} \rightarrow \{A\}$ C. $\{E\} \rightarrow \{F\}$ D. $\{A, E\} \rightarrow \{G\}$

   Shade X in the answer sheet if none of the options are true.

5. (2 points) Which of the following normal form is satisfied by R with Σ?

   A. 1NF B. 2NF C. 3NF D. BCNF

   Shade X in the answer sheet if none of the options are true.

Short Answer (24 points)

6. (8 points) Proof the following statement using the basic Armstrong's axioms: *Reflexivity*, *Augmentation*, and *Transitivity*. Use only the given number of lines.

   (a) (4 points) If $\{ X \rightarrow Y, (X \cup Y) \rightarrow Z \}$ then $\{ X \rightarrow Z \}$

   1. $X \rightarrow Y$ [Given]
   2. $(X \cup Y) \rightarrow Z$ [Given]
   3. ________________________________ [________________________________]
   4. ________________________________ [________________________________]

   (b) (4 points) If $\{ X \rightarrow Y, (Y \cup Z) \rightarrow W \}$ then $\{ (X \cup Z) \rightarrow W \}$

   1. $X \rightarrow Y$ [Given]
   2. $(Y \cup Z) \rightarrow W$ [Given]
   3. ________________________________ [________________________________]
   4. ________________________________ [________________________________]

7. (16 points) For this question, we will be using the following schema.

   - $R = A, B, C, D, E, F$
   - $\Sigma = \{ \{A, B\} \rightarrow \{C, D\},\ \{B, C, D\} \rightarrow \{A, F\},\ \{D, E\} \rightarrow \{A, F\},\ \{A, B, E\} \rightarrow \{D\},$
     $\{C, D, F\} \rightarrow \{A, B\} \}$

   (a) (3 points) Find all the candidate keys of R with Σ.

(b) (8 points) Consider the decomposition of R with Σ into $\delta = \{\{A, B, C, D\}, \{C, D, F\}, \{A, D, E, F\}, \{C, D, E\}\}$. Show that the decomposition is not a lossless-join decomposition by finding a *minimal* counterexample. You may use integers for all values. You do not have to use all the rows.

A	B	C	D

C	D	F

A	D	E	F

C	D	E

(c) (4 points) Find one lossless join **dependency preserving** decomposition of R with Σ in BCNF (*if any*). Otherwise, find one in 3NF.

(d) (1 point) Is your previous decomposition in BCNF or in 3NF?

Long Answer (16 points)

8. (16 points) For this question, we will be using the following schema.

- $R = A, B, C, D, E, F$
- $\Sigma = \{\{A\} \rightarrow \{C\}, \{B\} \rightarrow \{A\}, \{C\} \rightarrow \{A, B\}, \{D, E\} \rightarrow \{B\}, \{B\} \rightarrow \{E, F\}\}$

(a) (2 points) Find one canonical cover of $\Sigma = \{\{A\} \rightarrow \{C\}, \{B\} \rightarrow \{A\}, \{C\} \rightarrow \{A, B\}, \{D, E\} \rightarrow \{B\}, \{B\} \rightarrow \{E, F\}\}$ with respect to R using the algorithm introduced in the class.

(b) (4 points) Find one lossless-join dependency preserving decomposition of R with Σ in 3NF using the canonical cover computed in part (a) and using the algorithm introduced in the class.

(c) (4 points) Find **another** lossless-join dependency preserving decomposition of R with Σ in 3NF with the **fewest number of fragments**.

(d) (6 points) Create the table for the decomposition in part (c) enforcing all the functional dependencies in Σ. You may use `INT` for all type. Use the name of the attributes as the name of the table. For instance, if you have a table containing $\{A, B, C\}$, name the table `abc` (*all lowercase*).

Appendix

Armstrong's Axioms

- $\forall X \subseteq R : \forall Y \subseteq R : ((Y \subseteq X) \Rightarrow (X \rightarrow Y))$
  Rule: [Reflexivity]

- $\forall X \subseteq R : \forall Y \subseteq R : \forall Z \subseteq R : ((X \rightarrow Y) \Rightarrow ((X \cup Z) \rightarrow (Y \cup Z)))$
  Rule: [Augmentation of (#) with Z]

- $\forall X \subseteq R : \forall Y \subseteq R : \forall Z \subseteq R : (((X \rightarrow Y) \wedge (Y \rightarrow Z)) \Rightarrow (X \rightarrow Z))$
  Rule: [Transitivity of (#) and (#)]

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END OF PAPER